

Validation of Multi-contact elasto-plastic model using Finite Element Method

To validate the results discussed in Section 3.1, a particle structure with 3 particles under uniaxial compressive loading was simulated and compared against FEM results obtained using Ansys Mechanical.

In the FEM model, the particles were modelled as deformable bodies using a sphere of 10 mm diameter. An elasto-plastic constitutive law is applied to the particles. Frictionless contact interactions were explicitly defined using an Augmented Lagrange contact formulation. The 3 particles are inserted between 2 plates and a prescribed displacement of 5mm is applied to both the plates, such that they apply a uniaxial compression to the particle system. The parameters used for the FEM simulation are listed in Table S1.

Table S1: FEM Simulation parameters

Parameter	Value	Unit
Youngs Modulus	7.1e+10	Pa
Poisson ratio	0.33	-
Yield strength	2.8e+8	Pa
Mesh element size	6.6e-4	m

The DEM simulations were also performed under identical loading and geometrical conditions using the multi-contact elasto-plastic model and the Thornton-Ning model in Ansys Rocky. The material parameters used in the FEM simulations are used in the DEM simulations.

Validation is assessed by comparing force-displacement curves obtained from both methods. In figure S1, the results are demonstrated and show good agreement between FEM and DEM using the multi-contact elasto-plastic model throughout the whole compression regime. It also confirms that the multi-contact elasto-plastic model in DEM accurately captured the force response of an elasto-plastic deformation. Observed minor deviations are attributed to inherent differences in spatial discretization between mesh-based FEM and particle-based DEM approaches.

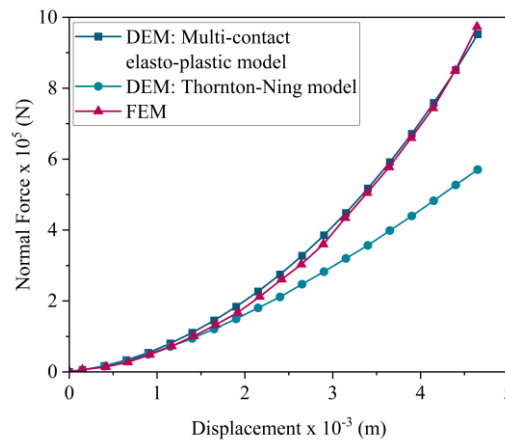


Figure S1: Force-displacement plots of a 3-particle system under compression